

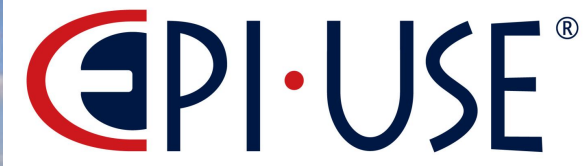
AutoFlow Trading

EPI-USE Africa

COS 301

Capstone Project Proposal
2026

Calvyn van Wyngaardt



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Investment, Real-time Data, Simulator, Virtual Transactions

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Project Aim

This project aims to provide a playground for students to simulate trades in a safe and virtual environment.



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Introduction

AutoFlow Trading is a simulation platform where users can learn how to trade in a safe sandbox environment. It functions on real-time market data and affords users with virtual currency to facilitate a realistic trading experience. Users can diversify their portfolio by investing in multiple different industries.



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Background & Motivation

Standard education lacks the hands-on experience needed to navigate real-world volatility safely. The goal in designing this trading platform is to enable users to learn trading by practical experience.

This project aims to test all the major pillars of software engineering. Students will be required to design databases, integrate multiple different APIs, visualize results in an interactive and compelling user interface, and potentially build AI agents.



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System Requirements



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Core Requirements

- Real Time (or Near Real Time) Data Visualization
 - Graph showcasing the price changes for a specific market share.
 - View price changes for different timeframes
 - Share price
 - Daily high & low
- Intuitive User Interface
 - Simple and easy to navigate
 - Show most popular investments
- User Portfolio Management
 - Virtual Account with initial funds
 - Current value of holdings
- Simulated Investments
 - Detailed log system for every trade
 - Ability to buy/sell immediately at the current market price
- Caching already fetched stock information (where allowed)



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Optional Requirements

- Multiple International Accounts
- Daily/Weekly report on Stock Changes
- Cryptocurrencies added
- Ability to buy/sell/notify when price hits a certain point
- Leaderboards



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Wow Factors [Optional]

- AI Portfolio investment assistant:
 - Recommend investment strategies based on current trends.
 - Predict potential price increases/decreases.
- Create a personalized report on the user's portfolio changes due to investment behavior with possible recommendations for improvement.
- Create an AI Agent to trade for the user
 - Include the ability to view Agent decisions, change maximum available funds for the Agent, etc.
 - Display the reasoning behind the decision (Optional).



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Architectural Requirements [Optional]



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Architecture Requirements

- The system shall implement secure payment protocols and methods for the virtual environment in a payment sandbox.
- The system shall be able to handle multiple concurrent changes effectively including multiple API requests and user transactions.
- The system must maintain 99.9% availability.
- The system must be able to support multiple concurrent transactions by distributing backend load.



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Design Requirements

- The system shall store personal user information in a relational database, with a user service responsible for personal information changes.
- The user must be able to see a graph of the stock price for a specific asset, indicating prices at multiple different points in time.
- The user must be able to switch between multiple currencies and stocks.
- There must be services responsible for querying and processing data from API services.



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Delivery Requirements [Optional]



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Delivery Requirements

- The application shall be delivered as a responsive web application and deployed to one of multiple free cloud hosting providers.

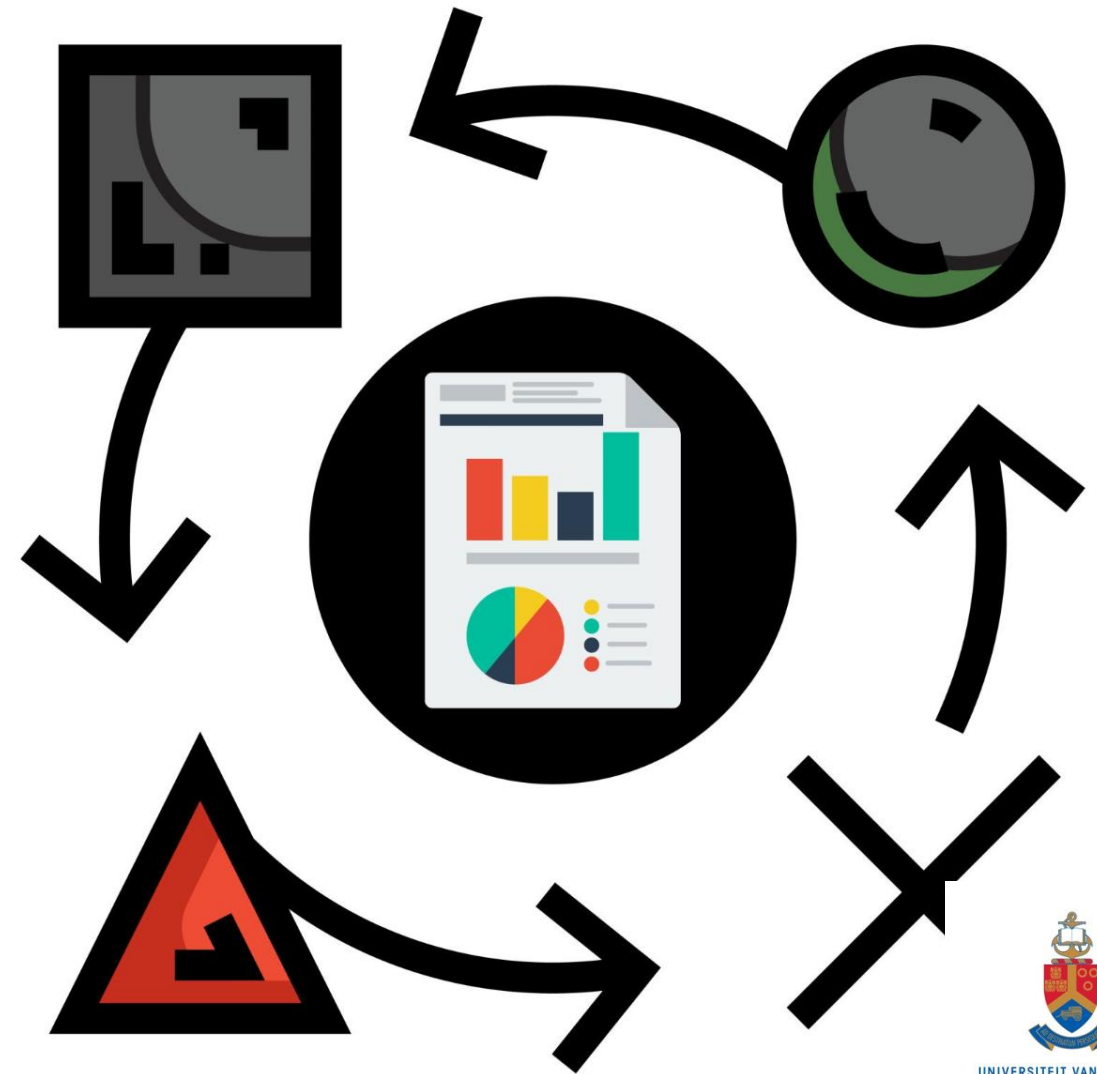


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Constraints [Optional]



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Constraints

- The system will be constrained by the maximum number of API calls allowed per minute.
- The system must not suffer from “slippage”.
- API Terms of Use must be followed when caching data
- No API calls to LLM's allowed. Students should build an agent from scratch for the WOW factor.



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Technology Stack



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Technology Stack

The final decision lies with the students, however the following tech stack is recommended:

- ReactJs + Tailwindcss
- Python FastAPI



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Budget



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Budget

EPI-USE will provide your team with a once-off amount of R1000 for the project's expenses. You are required to send a plan on how you will be spending the R1000.

My recommendation would be to first allocate funds for app deployment and then to the rest.

Some suggestions for utilizing the budget:

- App deployment
- T-shirts for the team
- Project Banners and posters for project day



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Client Details



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Contact Details

Project Owner

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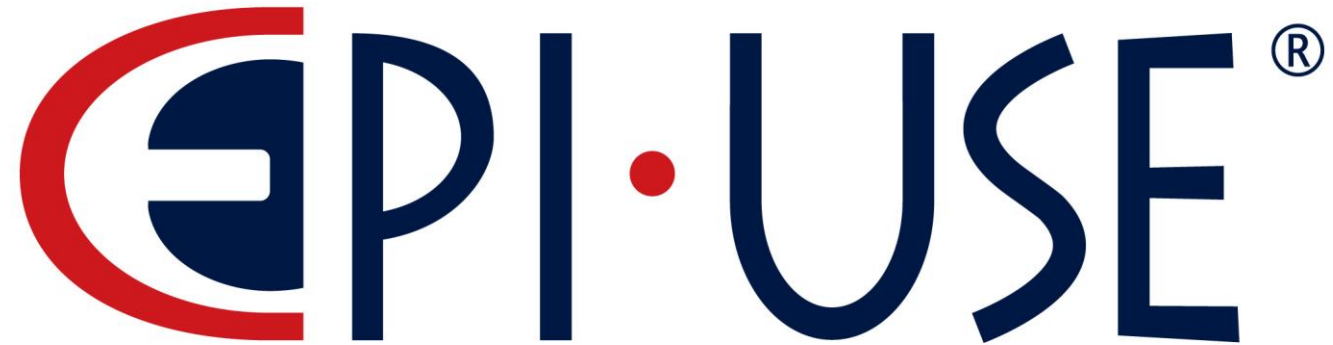
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Corporate Assets

Company Name: EPI-USE Africa (Pty) Ltd.

Address: 46 Ingersol Rd, Lynnwood Glen, Pretoria



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A photograph of the University of Pretoria's main building at night. The building is a large, historic stone structure with a prominent clock tower. It is illuminated with warm yellow lights, and the facade is decorated with colorful light projections in shades of blue, purple, and pink. A modern, multi-story building is visible in the background to the right. A large blue rectangle is overlaid on the left side of the image, containing the text 'THANK YOU' in white.

THANK YOU



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